

CRT - Answers are bolded

1.

a) List four legal identifier names.

rectangle , digits1, hurricane25, gradeavg

b) List four illegal identifier names and explain why each is illegal.

R3tangle Perimeter - has a random number, and a space

area@calc - @ is a illegal character

Total-Distance - uses the dash instead of underscore

Digits1&2 - has & and is uppercase

2.

a) In two statements, declare a variable named numBeads and assign it the value 5.

```
Int numBeads;  
numBeads = 5;
```

b) In one statement, declare a variable named numBeads and assign it the value 5.

```
Int numBeads = 3*10 / 6;  
or  
Int numBeads = 5;
```

3.

a) What is the final value of yourNumber after the last statement executes?

```
int myNumber = 5;
```

```
int yourNumber = 4;
```

myNumber = yourNumber * 2; (mn = 4*2 = 8)

yourNumber = myNumber + 5; (yn = 8+5=13)

13

b) What is the final value of yourNumber after the last statement executes?

int myNumber;

int yourNumber = 4;

myNumber = yourNumber + 7;

yourNumber = myNumber;

11

4. Determine the appropriate data type for each of the following values:

a) the number of basketballs in a department store.

Is an integer

b) the price of a basketball.

Float or decimal

c) the number of players on a basketball team.

int

d) the average age of the players on a basketball team.

int(probably) age cant be a half number so.

e) whether a basketball player has received a jersey or not.

True or false boolean

f) the first initial of a basketball player's first name.

A Character

5.

a) What is the difference between a primitive data type and an abstract data type?

Primitive data types store a single piece of data. Abstract data types store a set of data.

b) What is the difference between a class and an object?

A Class is like a template that can be used to define data, and a Object is something that is stored in the computer.

11. Using the following declarations, rewrite the statements to include the appropriate type casting, rounding where necessary. If type casting is not necessary, explain why:

```
int j = 5;  
double k = 1.6;  
int y;  
double z;
```

a) $y = j * k$;

```
Type.casting needed = true;
```

```
/*
```

*because multiplying an int by a double (k) will make the answer, which is an int into a double.

```
/*
```

b) $z = j * k$;

```
Type.casting needed = false;
```